MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TWISTED TRIAD, NO. 16 GAUGE, FLUOROPOLYMER INSULATION, FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 1TR-16

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction

First - Copper conductor, tin coated, AWG size 16, class B stranding.

Second - Insulation of ethylene tetrafluoroethylene (TEFZEL). Nominal wall thickness 0.010 inch. Maximum overall diameter 0.082 inch.

Third - Three insulated conductors, one black, one white, and one red, twisted together with a nominal lay of 2-1/2 inches to form a triad.

Fourth - Cable jacket of black polyether polyurethane applied so as to completely fill the triad valleys. The cable jacket material shall be B.F. Goodrich Co. ESTANE 58300 or 58863; Mobay Chemical Co. TEXIN 985A. Nominal wall thickness 0.100 inch. Overall diameter to be 0.380 to 0.400 inch.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - at 20°C, ohms/1000 feet, maximum.	4.24
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to water.....	2000
Insulation dielectric (on 100 percent of insulated conductor) - test in accordance with MIL-W-81381, using a test voltage of 8.0 kV (peak), minimum.	
Insulation resistance - megohms/1000 feet, minimum.....	5000

FSC 6145

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group A:	
Visual and dimensional	No failure
Group B:	
Cold bending, cable - at minus 40 ± 2°C over a 3-inch diameter mandrel	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum	5000
Elongation - percent, minimum	150
Cable jacket	
Tensile strength - lbf/in^2, minimum	5000
Elongation - percent, minimum	500
Group C:	
Physicals (aged)	
Insulation	
Air oven (168 hours at 180°C)	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	90
Cable jacket	
Air oven (168 hours at 80°C)	
This test shall be in accordance with MIL-C-915	No failure
Hot oil immersion	
Tensile strength, percent of unaged, minimum	90
Elongation, percent of unaged, minimum	80

QUALIFICATION INSPECTION:
Qualification inspection shall include basic electrical, plus all of groups A, B and C.

UNIT ORDERING LENGTH: 500 feet (minimum).

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Preparing activity:
Navy - SH
(Project 6145-0886-07)

Review activities:
Army - CR, ER
Navy - EC
DLA - IS

User activities:
Army - AL, AR, ME
Navy - CG

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